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METHOD FOR CONNECTING TETHER

Abstract

The present disclosure relates to a method for connecting a tether for anchoring a kite. The method for connecting a tether includes, while the kite is flying, separably connecting another tether different from one tether to the other end of the one tether whose one end is connected to the kite.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-035896 filed on Mar. 8, 2024, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to the technical field of methods for connecting a tether for anchoring a kite.

2. Description of Related Art

[0003] For example, connecting one fixing rope whose one end is connected to a submersible power generation platform and another fixing rope whose one end is connected to a waterway has been proposed as this type of method (see Japanese Unexamined Patent Application Publication (Translation of PCT Application) No. 2018-508712 (JP 2018-508712 A)).

SUMMARY

[0004] Part of a tether for anchoring a kite is often wound around, for example, a drum installed on the ground. The higher a target flight altitude of the kite, the longer the tether that anchors the kite, and therefore the larger the drum around which the tether is wound. As the drum becomes larger, a winch also becomes larger. A large winch is technically disadvantageous in its reduced response.

[0005] The present disclosure was made in view of, for example, the above issue, and it is an object of the present disclosure to provide a method for connecting a tether that can reduce a decrease in response of a winch.

[0006] A method for connecting a tether according to an aspect of the present disclosure includes while a kite is flying, separably connecting another tether different from one tether to another end of the one tether whose one end is connected to the kite.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0008] FIG. 1 is a diagram illustrating an example of a wind power generation system;

[0009] FIG. 2A is a diagram illustrating an example of a method of connecting tethers;

[0010] FIG. 2B is a diagram illustrating an example of a method of connecting tethers;

[0011] FIG. 3 is a flowchart illustrating an example of a method of connecting tethers;

[0012] FIG. 4 is a flowchart illustrating an example of a process of separating tethers; and

[0013] FIG. 5 is a diagram illustrating an example of a connection mode of the tether.

DETAILED DESCRIPTION OF EMBODIMENTS

[0014] An embodiment of a method for connecting tethers will be described with reference to FIGS. 1 to 5.

[0015] First, an outline of a tethered wind power generation system using a kite will be described with reference to FIG. 1. In FIG. 1, the wind power generation system includes a kite 1 and a ground equipment 2. The ground equipment 2 may comprise a winch having a drum wrapped with tethers for anchoring the kite 1. The ground equipment 2 may further comprise a generator connected to the axis of rotation of the drum. The ground equipment 2 moors the kite 1, and may be referred to as a mooring device. Note that the shape of the kite 1 shown in FIG. 1 is an example, and is not limited to this. In FIG. 1, although there is only one tether for mooring the kite 1, the kite 1 may be moored by a plurality of tethers.

[0016] In the wind power generation system, when the kite 1 rises, the tether is paid out from the winch drum as the kite 1 rises. The drum rotates due to the feeding operation of the tether. Then,

the generator rotates with the rotation of the drum, whereby power generation is performed. After the tether has been unwound to a predetermined length or a predetermined time has elapsed, the motor of the winch causes the drum to rotate in a direction that winds up the tether. As a result, the kite **1** is lowered due to the winding operation of the tether. In the wind power generation system, the power generation is performed by repeatedly performing the feeding operation and the winding operation of the tether.

[0017] Next, methods of connecting tethers for anchoring the kite **1** will be described referring to FIGS. **2A**, **2B**, and **3**. Here, a method of extending the tether for anchoring the kite **1** by connecting the tether **32** to the tether **31** for anchoring the kite **1** will be described.

[0018] In FIG. **2A**, the tether **31** extending from the drum **21** toward the kite **1** may pass through the lead portion **51**. One end of the tether **31** is connected to the kite **1**. Note that one end portion of the tether **31** may be directly connected to the kite **1**, or may be indirectly connected to the kite **1**. Here, “indirectly connected” may mean that the kite **1** and one end of the tether **31** are connected via another member (for example, another tether or the like). Note that the lead portion **51** and the lead portion **52** described later may be referred to as a fairlead.

[0019] A part of the tether **31** is wound around the drum **21**. At this time, the tether **31** may be wound around the drum **21** so that the other end **31a** of the tether **31** protrudes from the drum **21**. In the situation shown in FIG. **2A**, the rotating shaft of the drum **21** may be connected to a generator. The end **31a** of the tether **31** is placed in the connecting device **40**. When the length of the tether **31** wound around the drum **21** becomes shorter than the predetermined length, the end **31a** may be placed in the connecting device **40**. In other words, if the length of the tether **31** wound around the drum **21** is greater than the predetermined length, the end **31a** may not be placed in the connecting device **40**.

[0020] The tether **32** has an end **32a** and an end **32b**. The end **32a** of the tether **32** is placed in the connecting device **40**. A part of the tether **32** extending from the drum **22** toward the end **32a** may pass through the lead portion **52** and the plurality of guide pulleys. In other words, the end **32a** drawn from the drums **22** around which the tethers **32** are wound may be placed in the connecting device **40** through the lead portion **52** and the plurality of guide pulleys. As shown in FIG. **2A**, the tether **32** may be wrapped around the drum **22** such that an end **32b** of the tether **32** (i.e., an end opposite the end **32a** connected to the end **31a** of the tether **31**) protrudes from the drum **22**. The drums **21**, **22** may constitute a part of the ground equipment **2** shown in FIG. **1**. In the situation shown in FIG. **2A**, the rotating shaft of the drum **22** may not be connected to the generator.

[0021] The connecting device **40** may automatically connect the tether **31** and the tether **32**, for example, when the end **31a** of the tether **31** and the end **32a** of the tether **32** are placed in the connecting device **40**. For example, a connecting mechanism may be attached to each of the end **31a** of the tether **31** and the end **32a** of the tether **32**. Here, the connecting device **40** may connect the tether **31** and the tether **32** by fitting a connecting mechanism attached to the end **31a** and a connecting mechanism attached to the end **32a**. Examples of the connecting mechanisms include mechanical couplings and buckles. In this case, the connected tethers **31**, **32** can be separated by releasing the engagement state of the connecting mechanism. For example, no connecting mechanisms may be attached to the end **31a** of the tether **31** and the end **32a** of the tether **32**, respectively. Here, the connecting device **40** may connect the tether **31** and the tether **32** by twisting the vicinity of the end **31a** of the tether **31** and the vicinity of the end **32a** of the tether **32**. In this case, the twisted tethers **31**, **32** can be untwisted to separate the connected tethers **31**, **32**. In this manner, the connecting mechanism **40** may separably connect the tether **31** and the tether **32**.

[0022] After the tether **31** and the tether **32** are connected, the tethers **31** and **32** may be removed from the connecting device **40** and the guide pulley, for example. Thereafter, the portion of the tether **31** wound around the drum **21** is unwound from the drum **21**, whereby the tether **31** may be removed from the lead portion **51** and the drum **21**. Consequently, as shown in FIG. **2B**, the tether **32** extends from the drum **22** toward the kite **1**. In the condition shown in FIG. **2B**, the rotating

shaft of the drums **22** may be connected to a generator. In FIG. 2B, reference numeral “**33**” denotes a connection portion between the tether **31** and the tether **32**. By connecting the tether **32** to the tether **31**, the length of the entire tether anchoring the kite **1** is increased. Thus, the tether **32** may be referred to as an extension tether. A connecting mechanism may be attached to each of the end **32a** and **32b** of the tether **32**. In this case, it can be said that the extension tether has a connecting mechanism at both ends thereof.

[0023] A method of connecting the tethers will now be described with reference to the flowchart of FIG. 3. In FIG. 3, the end of one tether (e.g., the end **31a** of the tether **31**) is **S101** to the connecting device **40**. Other tether ends (e.g., the end **32a** of the tether **32**) are **S102** to the connecting device **40** in parallel with or in conjunction with **S101** process. The connecting device **40** then **S103** one tether and another tether.

[0024] In **S101** process, the end of one tether may be placed in the connecting device **40**, for example, by the connecting device **40** pulling the end of one tether into the connecting device **40**. Similarly, in the process of **S102**, the end of another tether may be placed in the connecting device **40**, for example, by the connecting device **40** pulling the end of another tether into the connecting device **40**. The connecting device **40** may connect two or more tethers to one tether.

[0025] The tether **31** and the tether **32** may be the same type of tether. The tether **31** and the tether **32** may be different types of tethers. For example, by connecting the tether **31** and the tether **32**, the flight altitude of the kite **1** can be increased. That is, the tether **31** is used at a higher altitude than the tether **32**. The higher the altitude, the lower the temperature, the more likely the tether (e.g., the moisture contained in the tether) is to freeze. In view of this, the tether **31** may be a tether that is more freeze-resistant than the tether **32**. That is, the tether **31** may be a freeze-resistant tether.

[0026] As described above, in the tether-type wind power generation system, in order to generate electric power, the feeding operation and the winding operation of the tether are repeatedly performed. As shown in FIG. 2B, when the tether **32** is connected to the tether **31**, the tether **32** is wrapped around the drum **22**. In this case, for power generation, the tether **32** is unwound from the drum **22**, and the tether **32** is wound onto the drum **22**. As a result, at least a portion of the tether **32** may be damaged by friction. In view of this, the tether **32** may be a tether with that is more abrasion-resistant than the tether **31**. That is, the tether **32** may be an abrasion-resistant tether.

[0027] A process of separating tethers will be described with reference to FIGS. 2A, 2B, and 4. Here, a method for shortening the tether for anchoring the kite **1** by separating the tethers **31**, **32** for anchoring the kite **1** will be described.

[0028] For example, the tether **32** may be wound on the drum **22** until the distance from the drum **22** (in other words, the ground equipment **2**) to the connection portion **33** reaches a predetermined distance in the condition shown in FIG. 2B. The tether **31** may then be wrapped around the drum **21**. As a result, the tension applied to the tether **31** due to the kite **1** is less likely to be applied to the connection portion **33**. Thereafter, the connection portion **33** may be placed in the connecting device **40**. If excessive force is not applied to the connection portion **33** and the connecting device **40**, the tether **31** may be wound around the drum **21** after the connection portion **33** is placed in the connecting device **40**.

[0029] The connecting device **40** may separate the end **31a** of the tether **31** constituting the connection portion **33** from the end **32a** of the tether **32**. For example, after the tether **31** and the tether **32** are separated, the axis of rotation of the drum **21** may be connected to a generator. Also, the drum **22** around which the tether **32** is wound may be removed from the ground equipment **2**.

[0030] A method for separating tethers will now be described with reference to the flowchart of FIG. 4. In FIG. 4, tethers anchoring the kite **1** are **S201** onto the drums **22**, for example. Thereafter, a tethered connection portion (e.g., connection portion **33**) is **S202** to the connecting device **40**. Before the processing of step **202**, a tether (e.g., tether **31**) closer to the kite **1** than the connection portion of the tether may be wound around a drum (e.g., drum **21**) different from the drum **22**. After the process of **S202**, the connecting device **40** separates one tether from another tether (**S203**).

Technical Effect

[0031] The higher the target flight altitude of the kite **1**, the longer the length of the tether anchoring the kite **1**. If the length of one tether is increased in order to achieve the target flight altitude of the kite **1**, the size of the drum around which the tether is wound also increases. On the other hand, in the present embodiment, the tether **32** is connected to the tether **31** that anchors the kite **1**, so that the length of the entire tether that anchors the kite **1** is increased. With this configuration, the length of each tether can be significantly shorter than the length for achieving the target flight altitude of the kite **1**. Therefore, the size of the drum (for example, at least one of the drums **21** and **22**) around which the tether (for example, at least one of the tethers **31** and **32**) is wound can be suppressed. Therefore, according to the present embodiment, deterioration of the response of the winch can be suppressed.

[0032] In the present embodiment, the number of tethers to be connected is not limited. Therefore, in order to reduce the size of the drum around which the tether is wound, the length of each tether may be shortened. That is, in the present embodiment, the drum may be miniaturized by actively shortening the length of each tether. In this case, the response of the winch can be improved due to the miniaturization of the drum. In addition, due to the miniaturization of the drum, the transportability of the drum (i.e., at least one of the drum on which the tether is wound and the drum on which the tether is not wound) can be improved.

[0033] Further, part of the plurality of connected tethers as tethers that anchor the kite **1** may be separated. In addition, instead of a separated tether, a new tether may be connected to the tether anchoring the kite **1**. That is, some tethers of the tether anchoring the kite **1** may be replaced. With this configuration, maintenance of the tether for mooring the kite **1** can be performed during flight of the kite **1**.

First Modification

[0034] A first modification of the tether connection method will be described with reference to FIG. **4**. As described above, two or more tethers may be connected to one tether that anchors the kite **1**. In the example shown in FIG. **5**, the tethers **32** and **34** are connected to the tether **31**. In this case, the connection portion **33** may mean a connecting portion between the tether **31**, the tether **32**, and the tether **34**. Here, the tether **34** may be a tether having higher conductivity than the tether **32**. The tether **34** may be grounded.

[0035] Depending on the weather conditions, lightning may strike the flight kite **1**. In this case, if no countermeasure is taken, there is a possibility that a current caused by lightning flows through the ground equipment **2** (see FIG. **1**) through the tether that moors the kite **1**. On the other hand, in the example shown in FIG. **5**, even if lightning strikes the kite **1** in flight, most of the current caused by lightning can be expected to flow to the tether **34**. Therefore, according to the first modification, the ground equipment **2** is less likely to be damaged by the current caused by the lightning. In this case, the tether **34** may be referred to as a lightning protection tether.

Second Modification

[0036] A second modification of the tether connection method will be described. The kite **1** according to the second modification may be a kite used in a tether-type wind power generation system or may be a kite not used in a tether-type wind power generation system. Here, as a kite that is not used in the tether-type wind power generation system, for example, a kite for photovoltaic power generation that is equipped with a solar panel is exemplified.

[0037] In the second modification, the sensor may be disposed at a connection portion between one tether and another tether (for example, the connection portion **33** between the tether **31** and the tether **32** shown in FIG. **2B**). Examples of the sensor include at least one of a temperature sensor, a humidity sensor, an altitude sensor, an atmospheric pressure sensor, a wind speed sensor, a wind direction sensor, and a GPS (Global Positioning System) sensor. The sensor disposed at the connection portion may be a radiosonde.

[0038] According to the second modification, for example, weather observation can be performed

by a sensor disposed at the connection portion between one tether and another tether. In addition, since the kite **1** can stay in the air for a relatively long period of time, it is possible to perform the fixed-point observation of the weather by the sensor.

[0039] Aspects of the disclosure derived from the above-described embodiments and modifications are described below.

[0040] A method for connecting a tether according to an aspect of the present disclosure includes, while a kite is flying, separably connecting another tether different from one tether to the other end of the one tether whose one end is connected to the kite.

[0041] The tether of the aspect may be a freeze-resistant tether, and another tether may be an abrasion-resistant tether. Another tether may be a lightning protection tether. Part of the one tether may be wound around the drum in such a that the other end protrudes from the drum, before the one tether is connection to another tether. A sensor may be disposed at a connection portion between the other end of the one tether and another tether.

[0042] The present disclosure is not limited to the above-described embodiments, and can be modified as appropriate within the scope and spirit of the disclosure that can be read from the claims and the entire specification, and a method of connecting a tether with such a modification is also included in the technical scope of the present disclosure.

Claims

1. A method for connecting a tether, the method comprising while a kite is flying, separably connecting another tether different from one tether to another end of the one tether whose one end is connected to the kite.
 2. The method according to claim 1, wherein: the one tether is a freeze-resistant tether; and the other tether is an abrasion-resistant tether.
 3. The method according to claim 1, wherein the other tether is a lightning protection tether.
 4. The method according to claim 1, wherein before the one tether is connected to the other tether, part of the one tether is wound around a drum in such a manner the other end of the one tether protrudes from the drum.
 5. The method according to claim 1, wherein a sensor is disposed at a connection portion between the other end of the one tether and the other tether.
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